

A 12V-to-1V 100A Inverted Pyramid Trans-Inductor Voltage Regulator Converter with 93.6% High Efficiency and Fast Transient Response

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Abstract

This paper proposes an inverted pyramid trans-inductor voltage regulator (IP-TLVR) converter that can offer significant benefits, including extended duty cycle, reduced power loss, and lower voltage stress. Additionally, it exhibits superior transient performance without additional complex controls by using trans-inductor (TL). This architecture has two inductor current outputs, which greatly increase the power density. This topology features allow the inductor current to be auto-balanced. It is manufactured using a 0.18 μ m BCD process, which achieves a peak efficiency of 93.6%, a maximum output current of 100A, and attains an 8 μ s recovery time for a 90A/1 μ s load transition.

I. Introduction

With the growth of data centers, the demand for high-power CPUs and GPUs is increasing (ranging from tens to hundreds of amperes). Multi-phase buck converters [1-2] are widely used. However, the conventional buck converter experiences notable efficiency drop due to its low duty ratio ($D < 0.1$) and high voltage stress on power MOSFETs. In recent years, double step-down (DSD) converter [3-6] architectures are popular since they can double the duty, enabling effective frequency increase at very low duty. Nevertheless, DSD still encounters the problem of high voltage stress. Thus, a quadruple step-down (QSD) converter [7] architecture in Fig. 1 was proposed to solve this problem. QSD has four times the duty of the conventional buck converter, and its power MOSFETs sustain lower voltage stress. However, this structure contains three flying capacitors, which need an extra circuit to balance the voltage, and the voltage conversion ratio (VCR) is limited to 0-1/16, failing to reach 12V-to-1V voltage conversion. Moreover, achieving a rapid transient response poses a challenge in high-current applications.

II. Proposed IP-TLVR Converter

The proposed IP-TLVR, as depicted in Fig. 2, employs three flying capacitors (C_{F1-3}) to reduce voltage stress. This enables the use of low voltage (LV) power MOSFETs, thereby enhancing power density and reducing the die area of power MOSFETs. The voltage stress of M_{1-4} is $V_{IN}/2$, and the voltage stress of M_{5-6} is $V_{IN}/4$. The numbers of high-side and low-side power MOSFETs are 4 and 2, respectively, which can spread the current evenly, reduce the conduction loss, and achieve better heat distribution. It can also extend the duty cycle to 1/3 in 12V-to-1V application, which is 4 times compare to that of the conventional buck converter. In addition, the proposed IP-TLVR replaces the original inductor with a TL, which generates a coupling current when a transient occurs, thereby increasing the slope of the inductor current to improve transient performance.

A. Proposed Trans-Inductor

Fig. 3 shows the proposed TL. Unlike the traditional inductor which has only two pins, the TL has an additional secondary side, enabling the TL to have four pins without increasing the volume. The secondary sides between different phases are connected in series to form a loop, so that current can be conducted on the secondary side of each phase.

B. Operation principle of proposed IP-TLVR in steady state

Fig. 4 shows the operating principle of the QSD converter. In steady state, $TD = 0$, the secondary side is open circuit, TL is equivalent to a general inductance, and the inductance value is L_m . In the operation of IP-TLVR, there are three states in one cycle. In state 1, M_1 , M_3 , and M_6 are turned on, where $V_{X1} = (V_{IN} - V_{CF1}) = (V_{CF2} - V_{CF3}) = 1/4 V_{IN}$ and $V_{X2} = 0$, TL_1 is magnetized and TL_2 is demagnetized. I_{TL1} is provided by two paths that flow through M_1 and M_3 . Meanwhile, C_{F1} is charged, C_{F2} is discharged, and C_{F3} is charged. In state 3, M_5 and M_6 are turned on. TL_1 and TL_2 are demagnetized, I_{TL1} and I_{TL2} flow

through LV MOSFET M_5 and M_6 , respectively, which can further reduce the conduction loss. In state 2, M_2 , M_4 , and M_5 are turned on. TL_1 is demagnetized and TL_2 is magnetized, and $V_{X1} = (V_{CF1} - V_{CF2}) = V_{CF3} = 1/4 V_{IN}$. I_{TL2} is provided by two paths that flows through M_2 and M_4 . Meanwhile, C_{F1} is discharged, C_{F2} is charged, and C_{F3} is discharged. These two paths can also reduce the conduction loss by half. The voltages on C_{F1-3} are $3/4V_{IN}$, $2/4V_{IN}$, and $1/4V_{IN}$, respectively.

C. Operation principle of proposed IP-TLVR in transient

As shown in Fig. 5, when $TD = 1$, M_{S1} and M_{S2} are turned on. The secondary side of TL begins to conduct, and the TL starts to operate. When the IP-TLVR converter starts working, in state 1, the voltage across TL_1 is $(1/4V_{IN} - V_{OUT})$. Since the primary-to-secondary turn ratio is 1:1, the secondary side voltage that corresponds to TL_1 will be the same. Similarly, since the secondary side cross voltage corresponding to TL_2 is about $-V_{OUT}$, the cross voltage on the compensation inductor L_C is $1/4V_{IN} - 2V_{OUT}$. This will cause a current I_{LC} to be generated on the secondary side and I_{LC} will flow through all the secondary sides connected in series, which will then be synchronously coupled to the primary side of all phases. Consequently, the slopes of I_{TL1} and I_{TL2} both rise in state 1 compared to the traditional inductor current, and simultaneously accelerate the response to the transient current. In state 3, the secondary side cross voltage corresponding to TL_1 and TL_2 is about $-V_{OUT}$, so the cross voltage on the compensation inductor L_C is $-2V_{OUT}$. Similarly, I_{LC} will also be coupled to the primary side, causing the absolute value of the inductor current slope to increase. The transient slew-rate is shown on the bottom left of Fig.5, where L_m is the mutual inductance of TL and L_r is the Leakage inductance of TL. Although the inductor current slope can be increased by the TL, meaning the inductor current ripple will increase, the proposed uses transient detect technology to turn off the secondary side when $TD = 0$ in steady state. When a load transient occurs, $TD = 1$, the secondary side turns on, enabling the inductor current slope to increase and the transient response to become faster.

III. Experimental Results

Fig. 6 shows the measured up and down transitions with conventional inductors and TLs, respectively. For the IP-TLVR, the V_{OUT} undershoot and overshoot voltages from 10A to 100A within 1 μ s load step are -180mV and 102mV with traditional inductors. However, the V_{OUT} undershoot and overshoot voltages are reduced to -48mV and 20mV, respectively, with TLs. Left of Fig. 7 shows the steady-state waveform of IP-TLVR. The inductor currents of the two phases are auto-balanced to be $I_o/2$ since the adaptive on-time control can achieve the same on-time in steady state. The voltage stress of low-side power MOSFETs are both $V_{IN}/4$, which enables the use of low-voltage MOSFET and reduces chip area. Middle of Fig. 7 shows the VCR versus duty cycle. Under the same VCR situation, IP-TLVR can have greater duty and has the widest range. Right of Fig. 7 shows the efficiency versus I_o in different V_{OUT} . The peak efficiency is 93.6% when $V_{OUT} = 1V$ and $I_o = 30A$. The chip micrograph uses a 0.18 μ m BCD process with 70%reduced power MOSFETs area, as shown in Fig. 8. Table I presents the performance summary and comparison table. The proposed design can achieve the lowest voltage stress, and the highest maximum I_o per phase.

References

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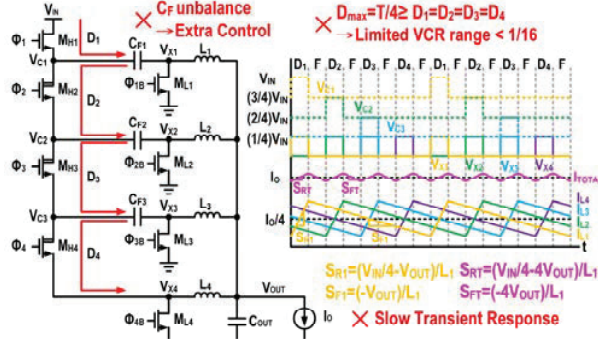


Fig. 1. QSD converter in [7] suffers from flying capacitor unbalance and slow transient response.

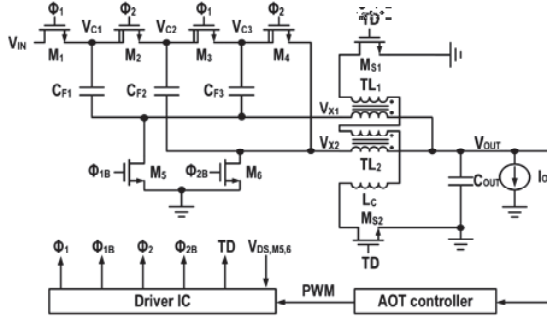


Fig. 2. Proposed 12V-to-1V inverted pyramid trans-inductor voltage regulator (IP-TLVR) converter.

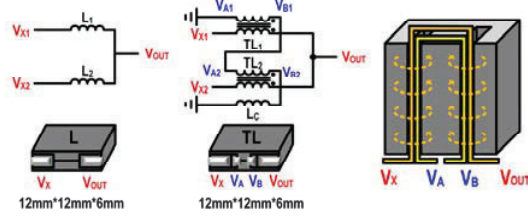


Fig. 3. Configuration of trans-inductor.

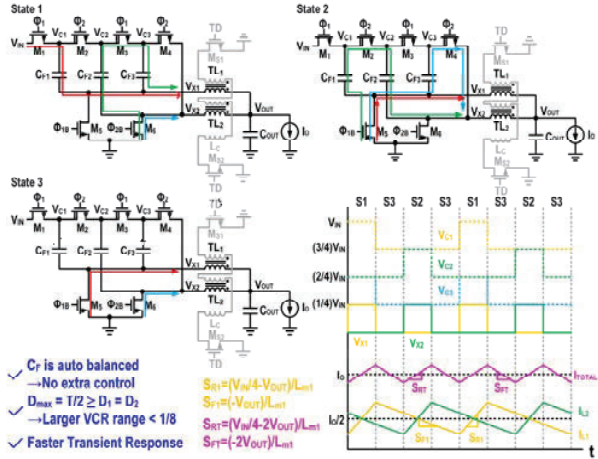


Fig. 4. Operation principle of proposed IP-TLVR (steady state).

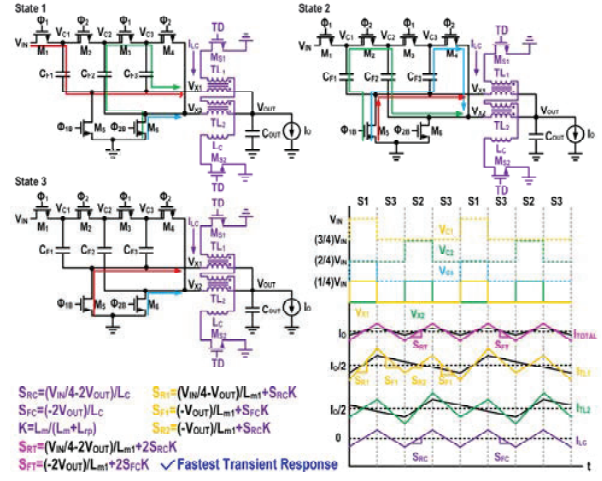


Fig. 5. Operation principle of proposed IP-TLVR (transient).

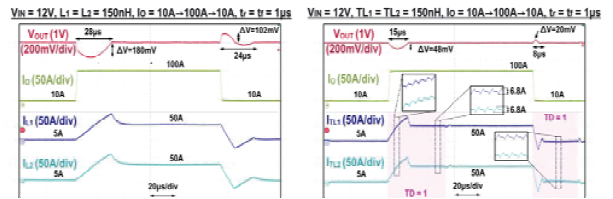


Fig. 6. Measured up and down transient response w/ conventional inductor and w/ trans-inductors.

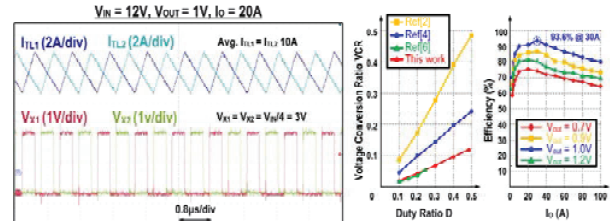


Fig. 7. Measured steady-state waveform (left), VCR versus duty ratio (middle), and efficiency versus I_o (right bottom).

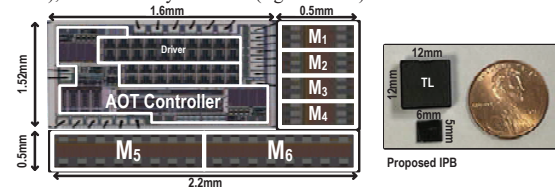


Fig. 8. Chip micrograph in the 0.18 μ m BCD process and photo of trans-inductor (TL) and proposed co-packaged IPB.

Table I. Comparison with state-of-the-art buck converters.

Publication	JSSC'2021 [2]	ISSCC'2022 [4]	ISSCC'2022 [5]	JSSC'2020 [6]	VLSI'2022 [7]	This Work
Process	0.6 μ m BiCMOS	0.18 μ m HV SOI CMOS	0.18 μ m BCD	0.18 μ m HV BCD	0.18 μ m BCD	0.18 μ m BCD
Topology	6-Phase Buck	CCC	DSD	DSD	QSD	IP-TLVR
Voltage Stress	V_{in}	$V_{in} \& 1/2V_{in}$	$V_{in} \& 1/2V_{in}$	$V_{in} \& 1/2V_{in}$	$1/2V_{in} \& 1/4V_{in}$	$1/2V_{in} \& 1/4V_{in}$
Input Voltage V_{in}	12V	12V	12V/24V	48V	48V	12V
Output Voltage V_{out}	1V	0.9V-1.8V	1V	1V	1V	1V
Maximum I_o	180A	4A	4A	1.5A	10A	100A
Switching Frequency	400KHz	2MHz	1MHz	100KHz-2MHz	500KHz/1MHz	1MHz
Inductor	6x0.25 μ H	2x0.74 μ H	2x1.8 μ H	2x0.9 μ H	4x1 μ H	2x0.15 μ H (TL)
Flying Capacitor C_f	No need	2x2.2 μ F	1x4.7 μ F	1x1 μ F	3x2.2 μ F	3x47 μ F
Output Capacitor C_o	6000 μ F	10.6 μ F	10 μ F	22 μ F	20 μ F	2000 μ F
Control Scheme	Current mode	Synchronized Hysteretic	Voltage-mode PWM	AOT-T	HCOT-AH	AOT
Peak Efficiency	92.5% @400KHz	86.6% @2MHz	88.3% @1MHz	85.4% @100KHz	88.5% @500KHz	93.6% @1MHz
Load Transient ΔI_o	100A	4A	3A	1A	6.3A	90A
Load Step Edge Time	1 μ s/1 μ s*	20ns/20ns	20ns/NA	NA/NA	20ns/NA	1 μ s/1 μ s
V_{DS} / V_{GS}	50mV/NA	110mV/210mV	50mV/NA	160mV/NA	80mV/NA	48mV/20mV
Recovery Time	50 μ s/NA	1.5 μ s/2 μ s	0.9 μ s/NA	8.2 μ s/NA	1 μ s/NA	15 μ s/8 μ s

* Effective value estimated.